TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 1

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line

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AIM

To investigate the significance of the characteristic impedance (Z_0) and propagation constant (γ) of a transmission line using its primary constants R, L, G and C, and to study the effect of resistance, inductance, frequency and capacitance.

SOFTWARE REQUIRED

SCILAB Version 6.1.1

THEORY / FORMULAE

Characteristic impedance:

$$Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$$

Propagation constant:

$$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$$

where α is the attenuation constant in neper/km, β is the phase constant in radians/km, and $\omega = 2\pi f$.

PROCEDURE

- Open SCILAB software.
- Open a new file.
- Enter the required values of R, L, G, C and frequency.
- Type and execute the SCILAB program.
- Record Z_0, α and β.
- Repeat the calculation for each variation specified in the test cases.

TEST CASE 1 – BASIC CALCULATION

Investigate Z_0 and γ for the given transmission-line primary constants.

Parameter	Value	Unit
R	10	Ω/km
L	3.5	mH/km
C	8	µF/km
G	4	µmho/km
f	2	kHz

TEST CASE 1 – RESULT TABLE

S.No.	Input	Z_o (Ω)	α (neper/km)	β (rad/km)	Status
1	R=10, L=3 mH, C=16 µF, G=4 µmho, f=2 kHz	14 − j2	0.3621	2.7769	Matches attached SCILAB output

TEST CASE 2 – EFFECT OF RESISTANCE AND INDUCTANCE

At f = 2.5 kHz, the resistance is varied from 10 Ω to 50 Ω in steps of 10 Ω while the other constants are kept fixed. The inductance is varied from 3.5 mH to 11.5 mH with R, G and C constant, as specified in the lab manual.

A. Variable Resistance (R): L = 3.5 mH/km, C = 8 µF/km, G = 4 µmho/km, f = 2.5 kHz

S.No.	R (Ω/km)	Z_o (Ω) – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	10	21.00 − j1.89	0.2381	2.6392	0.2381	2.6392
2	20	21.25 − j3.74	0.4707	2.6702	0.4707	2.6702
3	30	21.63 − j5.52	0.6935	2.7184	0.6935	2.7184
4	40	22.12 − j7.19	0.9042	2.7796	0.9042	2.7796
5	50	22.68 − j8.77	1.1023	2.8502	1.1023	2.8502

B. Variable Inductance (L): R = 10 Ω/km, C = 8 µF/km, G = 4 µmho/km, f = 2.5 kHz

S.No.	L (mH/km)	Z_o (Ω) – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	3.5	21.00 − j1.89	0.2381	2.6392	0.2381	2.6392
2	6.0	27.42 − j1.45	0.1824	3.4463	0.1824	3.4463
3	8.5	32.62 − j1.22	0.1534	4.0990	0.1534	4.0990
4	11.0	37.10 − j1.07	0.1349	4.6617	0.1349	4.6617

TEST CASE 3 – EFFECT OF FREQUENCY AND CAPACITANCE

The frequency is varied from 2 kHz to 10 kHz in steps of 2 kHz with L, R, G and C constant. Capacitance is varied from 8 µF to 16 µF in steps of 2 µF with R, L and G constant, as specified in the lab manual.

A. Variable Frequency (f): R = 10 Ω/km, L = 3.5 mH/km, C = 8 µF/km, G = 4 µmho/km

S.No.	f (kHz)	Z_o (Ω) – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	2	21.05 − j2.36	0.2376	2.1161	0.2376	2.1161
2	4	20.95 − j1.19	0.2387	4.2123	0.2387	4.2123
3	6	20.93 − j0.79	0.2389	6.3128	0.2389	6.3128
4	8	20.92 − j0.59	0.2390	8.4144	0.2390	8.4144
5	10	20.92 − j0.48	0.2390	10.5165	0.2390	10.5165

B. Variable Capacitance (C): R = 10 Ω/km, L = 3.5 mH/km, G = 4 μmho/km, f = 2 kHz

S.No.	C (μF/km)	Z_o (Ω) – Simulation*	α – Simulation*	β – Simulation*	α – Theory	β – Theory
1	8	21.05 − j2.36	0.2376	2.1161	0.2376	2.1161
2	10	18.83 − j2.11	0.2656	2.3659	0.2656	2.3659
3	12	17.19 − j1.93	0.2910	2.5917	0.2910	2.5917
4	14	15.91 − j1.79	0.3143	2.7994	0.3143	2.7994
5	16	14.88 − j1.67	0.3360	2.9927	0.3360	2.9927

SCILAB PROGRAM

```
clear;
clc;

// Given parameters
R = 10;             // Resistance in ohms/m
L = 0.003;          // Inductance in H/m
f = 2 * 10^3;       // Frequency in Hz
G = 4 * 10^(-6);    // Conductance in S/m
Cap = 16 * 10^(-6); // Capacitance in F/m

// Angular frequency
w = 2 * %pi * f;

// Series impedance
Z = R + %i * w * L;

// Shunt admittance
Y = G + %i * w * Cap;

// Characteristic impedance
Zo = sqrt(Z / Y);

Zo_R = real(Zo);
Zo_X = imag(Zo);

printf("Characteristic impedance:\n");
printf("Zo = %.4f + j(%.4f) ohms\n", Zo_R, Zo_X);

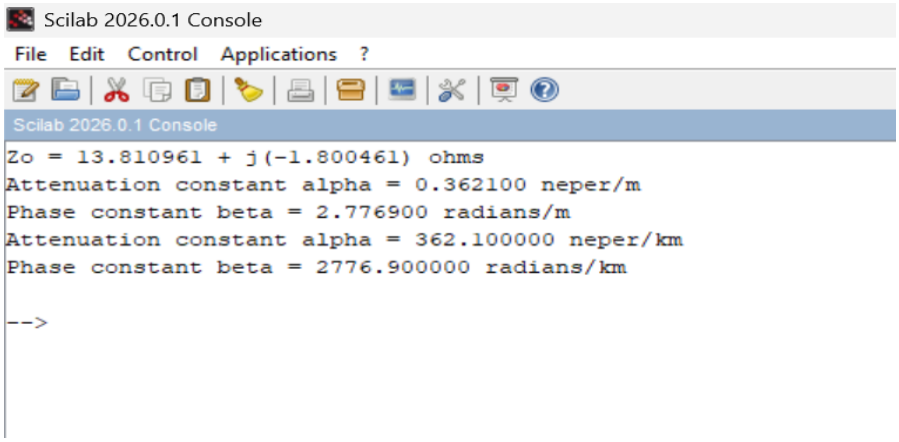
// Propagation constant
P = sqrt(Z * Y);

// Attenuation constant
alpha = real(P);

printf("\nAttenuation constant:\n");
printf("alpha = %.6f neper/m\n", alpha);
printf("alpha = %.4f neper/km\n", alpha * 1000);

// Phase constant
beta = imag(P);

printf("\nPhase constant:\n");
printf("beta = %.6f radians/m\n", beta);
printf("beta = %.4f radians/km\n", beta * 1000);
```



RESULTS

Test Case 1: The attached SCILAB output gives $Z_o = 14 - j2\ \Omega$, $\alpha = 0.3621$ neper/km and $\beta = 2.7769$ radians/km.

Test Case 2: The effect of resistance and inductance on Z_o, α and β was evaluated for the ranges specified in the manual.

Test Case 3: The effect of frequency and capacitance on Z_o, α and β was evaluated for the ranges specified in the manual.